DEBANJAN KAKATI

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EDUCATION

Birla Institute of Technology and Science, Pilani - Goa Campus Aug 2023 - Jul 2027
B.E. Chemical (Hons.) | Minor: Finance
Coursework: Portfolio Management, Derivatives & Risk, Valuation, Machine Learning, Deep Learning, Stochastic Processes

Gurukul Grammar Senior Secondary School 2010 - 2023
Class XII (CBSE): 90% | Class X: 91.6%

SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV), SQL, C++, JavaScript, MATLAB
AI & Quant: LLMs, RAG, Transformers, CNNs, GNNs, LSTM, Time-Series Forecasting, AR-GARCH, Monte Carlo, DCF
Tools: FastAPI, React, LangGraph, Docker, Azure, AWS, Power BI, Advanced Excel, PostgreSQL, Git, ANSYS, SolidWorks

EXPERIENCE

Digitalization Intern - Innovation & Learning Solutions May 2026 - Dec 2026
Shell India Markets Pvt. Ltd. *Bengaluru, India*

- Architected a digital platform for an **8-reactor** High-Throughput Experimentation (HTE) catalyst testing unit, cutting campaign data assembly from **~10 days to real time**.
- Built pipelines ingesting online **Gas Chromatography** and **SCADA** data, with mass balance closure.
- Automated **RON/MON** octane extraction and C1–C15 breakdowns from detailed hydrocarbon analysis reports.

Business Analytics & Industry 4.0 Intern May 2025 - Jul 2025
PIEDS - BITS Pilani (Govt. of India Society) *Remote*

- Evaluated **50+ early-stage startups** on market size, unit economics and growth potential to guide incubation decisions.
- Led outreach to **20+ venture capital funds**, building an investor and mentor network of **50+**.

Business & Operations Growth Associate Aug 2023 - Jul 2024
Intellipaat *Remote*

- Built **3 AI analytics systems** (decision platform, traffic optimization, HR automation), improving efficiency **30–45%**.
- Developed end-to-end data pipelines that **outperformed peer benchmarks by ~35%** in predictive accuracy.

PROJECTS

Ice Melting Rate Prediction via ML - Patent Filed | Prof. Anirban Roy

- Built a patent-pending ML model predicting ice melting rates with **96% accuracy** on **237+ experimental data points**.
- Benchmarked **12+ regression and deep learning models**; the tuned TensorFlow network needed **~40% less training data**.

Systemic Risk in the Cryptocurrency Ecosystem | Prof. Ashwini Kumar

- Combined **AR-GARCH**, **Transfer Entropy** networks and systemic risk indices across **8 digital assets over 10 years**.
- Showed stablecoins (USDT, USDC) **spread more systemic risk than BTC and ETH**, with quantile tail-risk models.

FreeWind - Air Quality Forecasting | Smart India Hackathon, National Top 1%

- Fused **Graph Neural Networks** and **LSTMs** over **300,000+ sensor records**, achieving **32% lower error** than baselines.

LEADERSHIP

SAE Formula Bharat - Chassis Development Engineer

- Delivered **7+ CAD/FEA designs** (ANSYS, SolidWorks) for a **Top 25 finish among 80+ teams**; early FEA saved **2+ weeks**.

PATENT, CERTIFICATIONS & ACHIEVEMENTS

- **Patent:** Ice Melting Rate Prediction Model, Indian Patent Office, App. No. IDF/H/115107 (filed Jan 2025, issued Nov 2025)
- **Certifications:** Advanced Data Science & AI, Intellipaat (2023); Advanced AI & Robotics, myEquation (2024)
- **Indian Space Academy:** Remote Sensing & GIS winter training